# A strictly positive biquadratic $M$-tensor without a sum-of-squares representation

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## Task 1: full resolution of the stated conjecture

Xu, Cui and Qi ask whether every symmetric biquadratic $M$-tensor is a sum of squares of bilinear forms [1, Question 1]. This was Conjecture 3.1 in their current preprint. We answer negatively, even with a strict $M$-eigenvalue margin.

Here symmetric means the tensor identities

$$a_{ijkl} = a_{kjil} = a_{klij}. \tag{1}$$

It does not impose invariance of the polynomial under arbitrary permutations of the $x$-variables or the $y$-variables. The source explicitly distinguishes these notions. We use exactly (1), its hypothesis in Question 1.

For $x, y \in \mathbb{R}^3$ define

$$C(x,y) = \sum_{i=1}^{3} x_i^2 y_i^2 + x_1^2 y_2^2 + x_2^2 y_3^2 + x_3^2 y_1^2 - 2\sum_{i<j} x_i x_j y_i y_j, \tag{2}$$

$$J(x,y) = \Big(\sum_{i=1}^{3} x_i^2\Big)\Big(\sum_{j=1}^{3} y_j^2\Big), \qquad P(x,y) = 8C(x,y) + J(x,y). \tag{3}$$

The form $C$ is the classical Choi form [2], up to cyclic orientation. We do not claim its PSD/non-SOS phenomenon as new. The point here is its application to the stated $M$-tensor question, with an explicit strict perturbation and a self-contained certificate.

**Theorem 1.** *The symmetric biquadratic tensor $\mathcal{A}$ of $P$ satisfies*

$$\mathcal{A} = 9\mathcal{I} - \mathcal{B}, \qquad \mathcal{B} \geq 0 \text{ entrywise}, \qquad \lambda_{\max}(\mathcal{B}) = 8 < 9.$$

*Thus $\mathcal{A}$ is a strong $M$-biquadratic tensor. Moreover, $P(x,y) \geq \|x\|^2\|y\|^2$, but $P$ is not a sum of squares of bilinear forms.*

### Positivity and the tensor entries

Let $\mathcal{T}$ be the tensor of $C$. Its nonzero entries are as follows. For $i = 1, 2, 3$, with $i+1$ taken cyclically,

$$t_{iiii} = t_{i,i+1,i,i+1} = 1.$$

For each $i < j$ set

$$t_{iijj} = t_{jiij} = t_{ijji} = t_{jjii} = -\tfrac{1}{2}.$$

Every other entry is zero. These prescriptions verify (1) and give $C = \sum_{i,j,k,l} t_{ijkl} x_i y_j x_k y_l$. The $M$-identity is $I_{ijkl} = \delta_{ik}\delta_{jl}$, whose form is $J$. Consequently $\mathcal{A} = 8\mathcal{T} + \mathcal{I}$ has integer entries.

**Lemma 1.** *$C(x, y) \geq 0$ for all real $x, y$.*

*Proof.* Write $C = x^T M(y) x$, where

$$M(y) = \begin{pmatrix} y_1^2 + y_2^2 & -y_1 y_2 & -y_1 y_3 \\ -y_1 y_2 & y_2^2 + y_3^2 & -y_2 y_3 \\ -y_1 y_3 & -y_2 y_3 & y_3^2 + y_1^2 \end{pmatrix}.$$

Put $a = y_1^2, b = y_2^2, c = y_3^2$. Its diagonal entries are nonnegative. The principal minors on index sets $12, 13, 23$ are, respectively,

$$ac + b^2 + bc, \qquad a^2 + ab + bc, \qquad ab + ac + c^2.$$

Its determinant is

$$a^2 c + ab^2 + bc^2 - 3abc \geq 0,$$

by AM–GM, since the product of the three positive terms is $(abc)^3$. The inequality includes zero values by continuity. All principal minors of the real symmetric matrix $M(y)$ are nonnegative, so it is positive semidefinite. This proves the lemma. □

Set $\mathcal{B} = 8(\mathcal{I} - \mathcal{T})$. Diagonal entries of $\mathcal{T}$ are zero or one, and its off-diagonal entries are zero or $-1/2$. Therefore $\mathcal{B}$ has entries in $\{0, 4, 8\}$ and is nonnegative. Its form is $8(J - C)$. If $\lambda$ is any $M$-eigenvalue of $\mathcal{B}$ with unit eigenvectors $x, y$, contraction with $x$ gives

$$\lambda = 8(1 - C(x, y)) \leq 8.$$

To verify attainment directly, set $v = (1, 1, 1)^T/\sqrt{3}$. For each fixed first index, the entries of $\mathcal{B}$ sum to 24; the same holds for each fixed second index. Hence

$$\mathcal{B} \cdot vvv = 8v, \qquad \mathcal{B}v \cdot vv = 8v.$$

Thus 8 is an $M$-eigenvalue and $\lambda_{\max}(\mathcal{B}) = 8$. This proves the asserted strict $M$-tensor condition. Lemma 1 also gives $P \geq J$.

## An exact obstruction to every SOS representation

Define a linear functional $\ell$ on the space of real biquadratic polynomials by specifying its values on monomials:

| monomial | $\ell$(monomial) |
|---|---|
| $x_i^2 y_i^2$ | 2 |
| $x_1^2 y_2^2,\ x_2^2 y_3^2,\ x_3^2 y_1^2$ | 1 |
| $x_2^2 y_1^2,\ x_3^2 y_2^2,\ x_1^2 y_3^2$ | 4 |
| $x_i x_j y_i y_j \quad (i < j)$ | 2 |

Its value on every unlisted monomial is zero. For an arbitrary bilinear form $g = \sum_{i,j=1}^{3} u_{ij} x_i y_j$, direct expansion gives

$$\begin{aligned} \ell(g^2) = {} & 2(u_{11} + u_{22} + u_{33})^2 + (u_{12} + 2u_{21})^2 \\ & + (u_{23} + 2u_{32})^2 + (u_{31} + 2u_{13})^2 \geq 0. \end{aligned} \tag{4}$$

In particular $\ell$ is nonnegative on every finite sum of squares of bilinear forms. On the other hand,

$$\ell(C) = 6 + 3 - 12 = -3, \qquad \ell(J) = 6 + 3 + 12 = 21, \qquad \ell(P) = 8(-3) + 21 = -3 < 0.$$

This contradicts any proposed SOS representation of $P$ and completes the proof of Theorem 1.

**Corollary 1.** *For every $0 < t < 1/7$, the form $C + tJ$ is strictly positive on pairs of nonzero vectors, its tensor is a strong $M$-biquadratic tensor, and it is not SOS.*

*Proof.* Its tensor is $(1 + t)\mathcal{I} - (\mathcal{I} - \mathcal{T})$, with $\lambda_{\max}(\mathcal{I} - \mathcal{T}) = 1$. Its form is at least $tJ$, while $\ell(C + tJ) = -3 + 21t < 0$. □

**Verification and scope.** The accompanying verifier checks all tensor entries and symmetries, all principal-minor identities, the exact eigenvector equations, and the separator identity for nine symbolic bilinear coefficients. These checks support the displayed algebraic proof; no numerical optimization, eigenvalue approximation, or sampled positivity claim is used. This resolves Question 1 only. Review is by the authoring agent, without independent peer review or a proof assistant.

## Four proposed follow-up tasks

1. Task 2: Determine the exact smallest $t \geq 0$ for which $C + tJ$ is SOS, and exhibit a representation at the threshold.
2. Task 3: Classify the SOS members of the three-parameter cyclic family with independently weighted diagonal and two cyclic square sums.
3. Task 4: Characterize which symmetric $M$-biquadratic tensors admit positive diagonal rescaling to a diagonally dominated tensor.
4. Task 5: Obtain dimension-dependent bounds for the amount of $M$-identity that must be added to a normalized $M$-biquadratic tensor to make it SOS.

These are proposed research tasks and earn no additional solution credit.

## References


[1] Y. Xu, C. Cui and L. Qi, *Sum of Squares Decompositions for Structured Biquadratic Forms*, Communications on Applied Mathematics and Computation (2026), doi:10.1007/s42967-026-00586-7. Preprint: arXiv:2512.02734v2.

[2] M.-D. Choi, *Positive semidefinite biquadratic forms*, Linear Algebra and its Applications **12** (1975), 95–100, doi:10.1016/0024-3795(75)90058-0.